Village Life Needs: A Comprehensive Framework for Nepal

Context: Rural Villages — Remote to Fully Connected

Contents

1 Introduction **4**

2 Water **4**
2.1 Drinking & Domestic 4
2.2 Irrigation 4
2.3 Irrigation Equipment & Machinery (Bare Minimum) 4
2.4 Emergency & Backup 5

3 Fire, Fuel & Energy **5**
3.1 Cooking Fuel 5
3.2 Electricity 5
3.3 Mechanical & Productive Use 5
3.4 Vehicular & Machine Fuel 6

4 Food **6**
4.1 Agriculture 6
4.2 Staples 6
4.3 Vegetables, Fruits & Other 6
4.4 Livestock 6
4.5 Storage & Processing 6
4.6 Markets & Distribution 7
4.7 Nutrition & Security 7

5 Shelter & Housing **7**
5.1 Construction 7
5.2 Comfort 7
5.3 Facilities 7
5.4 Resettlement 7

6 Clothing & Textiles **7**

7 Transportation **8**
7.1 No Motor Roads 8
7.2 With Motor Roads 8
7.3 Infrastructure 8

8 Education **8**
8.1 Early Childhood 8
8.2 School 8
8.3 Higher & Vocational 9
8.4 Adult & Non-Formal 9

8.5 Knowledge & Information . . . 9

9 Healthcare **9**
9.1 Primary . . . 9
9.2 Hospital & Referral . . . 9
9.3 Maternal & Child . . . 9
9.4 Specialized . . . 9
9.5 Traditional & Alternative . . . 9
9.6 Pharmacy & Drugs . . . 10

10 Administration & Government **10**
10.1 Offices . . . 10
10.2 Documentation . . . 10
10.3 Justice & Security . . . 10
10.4 Political & Civic . . . 10
10.5 Programs & Services . . . 10

11 Communication **10**
11.1 Traditional & Basic . . . 10
11.2 Telephony . . . 11
11.3 Internet & Digital . . . 11
11.4 Media . . . 11

12 Commerce, Finance & Markets **11**
12.1 Markets . . . 11
12.2 Finance . . . 11
12.3 Insurance . . . 11
12.4 Cooperatives . . . 11

13 Sanitation & Waste **11**

14 Social, Cultural & Religious **12**

15 Industry, Crafts & Livelihood **12**
15.1 Cottage & Handicraft . . . 12
15.2 Processing & Manufacturing . . . 12
15.3 Livelihood . . . 12

16 Environment & Natural Resources **12**

17 Resource Conservation **12**
17.1 Food Conservation . . . 13
17.2 Energy Conservation . . . 13
17.3 Water Conservation . . . 13
17.4 Material & Other Resource Conservation . . . 13
17.5 Community-Level Conservation . . . 13

18 Security & Emergency **14**

19 Vulnerable Groups & Social Protection **14**

20 Technology & Repair **14**

21 Public Infrastructure **14**

22 Legal & Rights **14**

23 Identity & Belonging **14**

24 100% Fully Independent Village: Complete Self-Sufficiency Checklist **15**
24.1 Water Independence . . . 15
24.2 Energy Independence . . . 15
24.3 Food Independence . . . 15
24.4 Shelter Independence . . . 15
24.5 Clothing & Textile Independence . . . 16
24.6 Transport Independence (Internal) . . . 16
24.7 Education Independence . . . 16
24.8 Healthcare Independence . . . 16
24.9 Governance & Administration Independence . . . 16
24.10 Justice & Security Independence . . . 16
24.11 Communication Independence . . . 17
24.12 Commerce & Finance Independence . . . 17
24.13 Sanitation & Waste Independence . . . 17
24.14 Industry, Craft & Repair Independence . . . 17
24.15 Raw Materials & Hard Dependencies . . . 17
24.16 Population & Demographics . . . 17
24.17 Social, Cultural & Religious . . . 17
24.18 Environment & Sustainability . . . 18
24.19 Time, Calendar & Measurement . . . 18
24.20 Surplus, Buffer & Risk . . . 18
24.21 Geographic & Resource Reality . . . 18
24.22 Skill Redundancy & Succession . . . 18
24.23 Death, Ritual & Continuity . . . 18
24.24 Conflict, Crime & Punishment . . . 19
24.25 Defense & External Threat . . . 19
24.26 Healthcare: The Limits . . . 19
24.27 Population & Demographics: The Numbers . . . 19
24.28 Social Obligation & Reciprocity . . . 19
24.29 Division of Labor: Caste, Gender . . . 19
24.30 Childcare, Elderly & Disability . . . 19
24.31 Storage & Preservation . . . 19
24.32 Pollution & Waste . . . 20
24.33 Beauty, Meaning & Spirit . . . 20
24.34 The Hard Truths . . . 20
24.35 Summary: Independence vs. Reality . . . 20

25 Summary: The Continuum **20**

26 References and Further Reading **21**
26.1 Infrastructure & Wellbeing . . . 21
26.2 Water & Irrigation . . . 21
26.3 Food Security & Post-Harvest . . . 21
26.4 Energy . . . 22
26.5 Health & Education . . . 22
26.6 Rural Development & Transport . . . 22

1 Introduction

This document catalogs the essential needs for village life in Nepal, considering the vast spectrum of conditions across the country. Nepal's topography—from the high Himalayas to the Terai plains—and varying levels of development mean that villages range from **fully remote** (no motor roads, no electricity, often accessible only by foot or mule) to **fully connected** (roads, grid electricity, internet). Each category below considers both extremes and the continuum between them.

Think large. This framework aims to be exhaustive—covering not only survival but dignity, not only basics but aspiration, not only individuals but community, not only present but future.

This document draws on research articles, policy reports, and documented studies on rural Nepal and village development (see References and Further Reading).

2 Water

2.1 Drinking & Domestic

- Springs (*kuwa*, *dhara*), wells, rivers, streams
- Rainwater harvesting, rooftop collection
- Piped supply, gravity-fed systems
- Water purification: boiling, filters, chlorine, UV
- Storage: tanks, *ghagri*, *gagri*
- Traditional: *ghaite* (stone spouts), *hitri*
- Water user committees, maintenance
- Seasonal scarcity: dry season, monsoon turbidity
- Water quality testing

2.2 Irrigation

- Canals (*kulo*), diversion weirs
- Lift irrigation, pump irrigation
- Drip, sprinkler (where feasible)
- Traditional *pani ghatta* (water mills)
- Pond, reservoir
- Snowmelt, glacier-fed (high altitude)

2.3 Irrigation Equipment & Machinery (Bare Minimum)

To make irrigation easier—especially where gravity-fed canals are insufficient or land needs preparation—villages need at least the following.

Water lifting & conveyance:

- **Pump:** Diesel, electric, or solar—to lift water from well, river, stream, pond when source is below field
- **Pipes & hose:** PVC, HDPE, flexible hose—to convey water from source to field; suction and delivery pipes
- **Hand/treadle pump:** Where no fuel or electricity—human-powered lift

Land preparation & earthwork:

- **Power tiller (2-wheel tractor):** Plowing, tilling, land leveling—levels field for even water distribution, reduces runoff; common in Nepal Terai and accessible hills

- **Tractor (4-wheel):** Where terrain and road allow—plowing, tilling, pulling trailer, sometimes pump; heavier work
- **Attachments:** Plow, cultivator, leveler, ridger—for tiller/tractor
- **Shovel, spade, *kodalo*:** Manual canal digging, repair—always needed; bare minimum when no machinery

Canal & distribution:

- **Canal lining:** Cement, plastic sheet—reduces seepage, saves water
- **Gates, valves:** To control flow, allocate water
- **Drip kit:** Main line, laterals, emitters—efficient for vegetables, orchards; saves water
- **Sprinkler:** Pipes, sprinkler heads—for larger area; needs pressure

Context:

- **No road:** Tractor/tiller cannot reach; manual tools, solar pump (portable), or gravity only
- **With road:** Power tiller, tractor, diesel pump—all feasible; shared or cooperative ownership common; FMIS (farmer-managed irrigation) accounts for 2/3 of Nepal's irrigated area and often outperforms agency-managed systems (see References)
- **With electricity:** Electric pump, solar pump—no fuel dependency
- **Maintenance:** Spare parts, repair skill—pump and machinery need upkeep

2.4 Emergency & Backup

- Hand pumps, tube wells
- Community water points
- Emergency tankers (disaster)

3 Fire, Fuel & Energy

3.1 Cooking Fuel

- Firewood: primary in remote areas; sustainable harvesting, community forests
- Improved cookstoves (*improved chulo*), smoke hoods
- Biogas: where livestock exists
- LPG/gas cylinders: where roads allow delivery
- Kerosene: backup; lanterns, stoves
- Solar cookers: experimental
- Electric stoves: where reliable electricity exists
- Agricultural residue: straw, dung cakes

3.2 Electricity

- **Off-grid:** Solar home systems (SHS), micro-hydro, pico-hydro— 3,300 micro-hydro plants in rural Nepal; research ranks micro-hydro best for rural electrification, then solar (see References)
- Solar lanterns, solar charging stations
- Grid connection (often unreliable)
- Micro-grid, mini-grid
- Backup: generators (diesel), batteries
- Street lighting
- Community charging points

3.3 Mechanical & Productive Use

- Grinding mills (*ghatta*), rice hullers

- Irrigation pumps
- Welding, carpentry power tools
- Refrigeration, cold storage
- Human & animal power (where no electricity)

3.4 Vehicular & Machine Fuel

- Petrol, diesel (where roads exist)
- Kerosene for generators
- Electric vehicles, charging (emerging)

4 Food

4.1 Agriculture

- Seeds: local, improved, hybrid; seed banks—research in mid-hills Nepal shows improved cultivars and crop yields as primary drivers of household food self-sufficiency (Karki et al., 2015; see References)
- Fertilizers: organic (compost, manure), chemical (urea, DAP)
- Pesticides, herbicides
- Tools: plough (*halo*), sickle (*hasiya*), *kodalo*, *khurpi*, spade
- Terraces, land leveling
- Agricultural extension, training
- Weather information, crop calendar
- Pest control, crop protection

4.2 Staples

- Rice, wheat, maize, millet, barley, buckwheat
- Varies by altitude: Terai (rice), hills (maize, millet), high hills (buckwheat, potato)
- Pulses, legumes

4.3 Vegetables, Fruits & Other

- Kitchen gardens, seasonal produce
- Fruit trees: citrus, mango, banana, apple (high altitude)
- Spices, herbs
- Storage: drying, pickling, smoking

4.4 Livestock

- Buffalo, cows, goats, sheep, chickens, pigs, ducks
- Milk, meat, eggs, manure, draft power
- Fodder, grazing land
- Veterinary services, vaccination
- Beekeeping, fish farming (ponds)

4.5 Storage & Processing

- Grain storage (*bharako*), rodent protection
- Cold storage (where electricity exists)
- Rice mills, oil mills, flour mills
- Cheese, ghee, *chyurpi* (dried cheese)

- Local processing vs. distant

4.6 Markets & Distribution

- Local shops, *haat bazaar*, weekly markets
- Access to district headquarters, wholesale
- Fair price, exploitation
- Food preservation for off-season

4.7 Nutrition & Security

- Dietary diversity
- Fortified foods, iodized salt
- School meals, mid-day meal
- Emergency food, relief
- Community kitchens (disaster)
- Malnutrition programs

5 Shelter & Housing

5.1 Construction

- Materials: stone, mud, wood, bamboo, brick, tin roofing
- Earthquake-resistant construction
- Retrofitting of old houses
- Ventilation, chimney (smoke)
- Insulation (cold areas)

5.2 Comfort

- Fireplace (*chulo*), warmth
- Blankets, bedding
- Lighting: sunlight, fire, kerosene, solar, electric

5.3 Facilities

- Toilet: pit latrine, pour-flush, septic
- Bathing area
- Kitchen, storage

5.4 Resettlement

- Landslide, flood relocation
- Planned resettlement
- Temporary shelter (disaster)

6 Clothing & Textiles

- Fabric, cloth; local weaving vs. market
- Tailoring, sewing, repair
- Seasonal: winter, monsoon, summer
- Traditional wear: *daura suruwal*, *sari*, *gunyu*
- Footwear: sandals, shoes, boots

- School uniforms
- Washing, drying

7 Transportation

7.1 No Motor Roads

- Foot trails, *ghat* (stone steps)—research finds road access is the highest infrastructure priority in rural Nepal; remote areas show 50–65% lower per capita income (see References)
- Trail maintenance, landslide clearance
- Pack animals: mules, horses, yaks (high altitude)
- Porters: human carriers
- Ropeways: gravity goods (*tatopani*), passenger
- Rivers: boats, rafts, canoe (Terai, hills)
- Bridges: suspension, wooden, rope
- Helicopter: emergency only

7.2 With Motor Roads

- Buses, jeeps, trucks, tractors
- Motorcycles, bicycles
- Ambulance, emergency vehicles
- Goods delivery
- Bus stops, parking

7.3 Infrastructure

- Road construction, maintenance
- Culverts, drainage
- Retaining walls
- Ferry services (rivers)
- STOL airstrips (remote)

8 Education

8.1 Early Childhood

- Daycare, *bal bikas* centers
- ECD (Early Childhood Development)
- Mothers' groups, community-run

8.2 School

- Primary (1–5), secondary (6–10), higher secondary (11–12)
- Buildings, classrooms, toilets, drinking water
- Teacher availability, training, housing
- Multi-grade teaching (remote)
- Textbooks, school supplies, uniforms
- Mid-day meal
- Girls' education, menstrual hygiene
- School management committee

8.3 Higher & Vocational

- Colleges, universities
- Vocational: carpentry, tailoring, agriculture, plumbing, electrician
- Distance learning (internet)
- Scholarships, hostel

8.4 Adult & Non-Formal

- Literacy classes
- Farmer field schools
- Women's groups, cooperatives

8.5 Knowledge & Information

- Library, reading materials
- Bulletin boards, announcements

9 Healthcare

9.1 Primary

- Health posts, sub-health posts
- FCHV (Female Community Health Volunteers)
- Basic medicines, first aid
- Vaccination

9.2 Hospital & Referral

- District, zonal, regional hospitals
- Kathmandu for specialized care
- Ambulance, doko, stretcher, helicopter
- Lab, X-ray, surgery

9.3 Maternal & Child

- Antenatal care, delivery (home vs facility)
- Birthing centers
- Nutrition, immunization

9.4 Specialized

- Dental, eye care (often camp-based)
- Mental health (underserved)
- Disability, rehabilitation
- Elderly, palliative care

9.5 Traditional & Alternative

- *Dhami/jhankri*, Ayurveda
- Herbal medicine

9.6 Pharmacy & Drugs

- Drug availability, quality
- Pharmacy, health post stock

10 Administration & Government

10.1 Offices

- Village/municipal office (*gaunpalika/nagarpalika*)
- Ward office (*ward karyalaya*)
- Ward chairperson, staff
- District office access

10.2 Documentation

- Citizenship (*nagarikta*)
- Birth, death, marriage registration
- Land ownership (*lalpurja*), registration
- Tax payment
- PAN, business registration
- Passport, visa

10.3 Justice & Security

- Police post
- Court access
- Mediation, dispute resolution
- Domestic violence, child protection

10.4 Political & Civic

- Election booths, voter registration
- Public hearings, participation
- Citizen assembly, planning

10.5 Programs & Services

- Census, surveys
- Disaster relief
- Employment programs
- Subsidies: fertilizer, seeds, elderly, disabled, single women
- Food security programs

11 Communication

11.1 Traditional & Basic

- Postal service
- Radio (FM, Nepal Radio)
- Word of mouth, messengers
- Bulletin boards

11.2 Telephony

- Mobile (2G, 3G, 4G)
- Landline (rare)
- Satellite phone (emergency)

11.3 Internet & Digital

- Mobile data
- Broadband, fiber (rare)
- Cyber cafes, community internet
- Smartphones, social media
- E-governance, online services
- Digital payments, mobile banking

11.4 Media

- Television, cable, satellite
- Newspapers (where available)

12 Commerce, Finance & Markets

12.1 Markets

- Local shops, *haat bazaar*
- Wholesale, retail
- Weekly, seasonal markets

12.2 Finance

- Banks, branches, mobile banking
- Cooperatives, savings groups
- Microfinance
- Credit, loans
- Remittance transfer

12.3 Insurance

- Life, health, property
- Crop, livestock (emerging)

12.4 Cooperatives

- Agricultural: input supply, marketing
- Consumer
- Credit

13 Sanitation & Waste

- Toilets: pit, pour-flush, septic; open defecation
- Waste: composting, burning, dumping
- Drainage, flooding
- Solid waste management (rare)

14 Social, Cultural & Religious

- Temples, *gumbas*, mosques, churches
- Community halls (*samaj ghar*)
- Festivals, *jatras*, ceremonies
- Cemeteries, cremation grounds
- Sports: football, volleyball, cricket, *dandi biyo*
- Youth clubs, women's groups
- Music, dance, theater
- Cinema (where available)

15 Industry, Crafts & Livelihood

15.1 Cottage & Handicraft

- Weaving, knitting
- Pottery, metalwork
- Bamboo, cane work
- Handicraft for market

15.2 Processing & Manufacturing

- Rice mills, oil mills
- Sawmills, brick kilns
- Cheese, ghee, jam
- Small-scale manufacturing

15.3 Livelihood

- Agriculture, livestock
- Wage labor, seasonal migration
- Remittance (major)
- Business, shopkeeping
- Government, NGO employment
- Tourism: homestay, trekking, guide

16 Environment & Natural Resources

- Forests: community, government; fuel, timber
- Grazing land
- Soil conservation, landslide prevention
- Biodiversity, wildlife
- Climate adaptation
- Seasonal calendar: monsoon, drought, cold

17 Resource Conservation

Conserving food, energy, water, and other resources reduces waste, extends availability, and builds resilience—especially where supply is limited or seasonal.

17.1 Food Conservation

- **Cold storage:** Community cold store (electricity or solar-powered) for vegetables, fruits, dairy, meat—extends shelf life, reduces post-harvest loss (20–50% for horticulture without it; see References), enables off-season supply
- **Evaporative cooling:** *Pot-in-pot*, sand storage—low-tech cooling without electricity
- **Drying:** Sun-drying (*sukeko*), solar dryer—grains, vegetables, fruits, fish, meat (*sukuti*); reduces moisture, prevents spoilage
- **Pickling & fermentation:** *Achar*, *gundruk*, *sinki*, *kinema*—preserves vegetables, extends use
- **Smoking:** Meat, fish—traditional preservation
- **Salt curing:** Fish, meat
- **Grain storage:** *Bharako* (elevated, rodent-proof), neem leaves, ash—protects from pests, moisture
- **Seed storage:** Cool, dry, sealed—maintains viability
- **Root cellar:** Underground or shaded—potato, onion, ginger
- **Reducing waste:** Proper handling, timely harvest, first-in-first-out

17.2 Energy Conservation

- **Cooking:** Improved cookstoves (less fuel), pressure cooker, lid on pot, batch cooking
- **Lighting:** LED bulbs, solar lanterns—efficient use of limited power
- **Insulation:** Houses—reduce heating/cooling need
- **Appliance efficiency:** Energy-efficient fridge, fan, pump—where electricity exists
- **Peak avoidance:** Use mills, pumps during off-peak—if grid with load management
- **Biomass:** Sustainable harvest, replanting—don't deplete forest
- **Waste heat:** Biogas slurry for fertilizer; improved chulo heat for water

17.3 Water Conservation

- **Rainwater harvesting:** Rooftop, tanks—capture monsoon
- **Greywater reuse:** Washing water for garden, livestock
- **Irrigation:** Drip, sprinkler—reduce loss; canal lining
- **Leak repair:** Pipes, taps—prevent waste
- **Spring protection:** Source conservation, recharge
- **Storage:** Tanks, ponds—for dry season

17.4 Material & Other Resource Conservation

- **Reuse:** Containers, bags, tools—repair before replace
- **Recycling:** Metal, plastic—where collection exists; composting organic
- **Timber:** Sustainable harvest, efficient use, alternative (bamboo)
- **Soil:** Compost, mulch, terrace—conserve fertility
- **Medicines:** Proper storage, expiry awareness—avoid waste
- **Fuel:** Efficient stoves, no idle burning

17.5 Community-Level Conservation

- **Grain bank:** Collective storage, buffer for lean season
- **Seed bank:** Preserve local varieties, share
- **Water user group:** Fair allocation, maintenance
- **Forest user group:** Regulated harvest, regeneration

18 Security & Emergency

- Natural disasters: earthquake, landslide, flood
- Early warning, evacuation
- Fire: limited services; community response
- Conflict, dispute
- Wild animals, snakes
- Search & rescue
- Relief distribution

19 Vulnerable Groups & Social Protection

- Elderly care, pension
- Orphan, child protection
- Disabled support
- Widows, single women
- Dalit, marginalized
- Indigenous, ethnic
- Refugees (if applicable)

20 Technology & Repair

- Mobile repair
- Battery charging
- Solar installation, repair
- Agricultural machinery repair
- Bicycle, motorcycle repair

21 Public Infrastructure

- Roads, bridges
- Street lighting
- Public spaces, playgrounds
- Warehouses, storage
- Parking, bus stops

22 Legal & Rights

- Marriage, divorce
- Inheritance, property
- Labor rights
- Child rights
- Women's rights
- Land rights, tenancy

23 Identity & Belonging

- Caste, ethnicity
- Language, mother tongue
- Religion

- Voter ID
- Community membership

24 100% Fully Independent Village: Complete Self-Sufficiency Checklist

What must a village have *within its boundaries* to be 100% fully independent—requiring no external inputs, services, or markets for survival and functioning? This section catalogs every need for maximum self-reliance. Some items are aspirational; some have **hard dependencies** (noted) that may require minimal external trade or are nearly impossible to produce locally.

24.1 Water Independence

- Own water source: spring, well, river, or sufficient rainwater harvesting
- Own gravity-fed or locally pumped distribution system
- Own purification: boiling, sand filter, solar disinfection—no external chemicals
- Own irrigation: *kulo*, ponds, lift from local stream
- Own maintenance capacity: masons, plumbers
- Water user committee for governance

24.2 Energy Independence

- **Cooking:** Firewood from community forest (sustainably managed), biogas from livestock, agricultural residue—no LPG, kerosene
- **Electricity:** Micro-hydro, pico-hydro, or solar—no grid dependency
- **Mechanical:** Water mills (*ghatta*), animal power, human power—or electric mills from local generation
- **Lighting:** Solar, fire, oil from local seeds (mustard, etc.)
- **Hard dependency:** Solar panels, batteries, micro-hydro turbines—typically imported; for true independence would need local manufacture or long-life equipment

24.3 Food Independence

- **Seeds:** Own seed saving, community seed bank—no hybrid dependency
- **Fertilizer:** Compost, manure, green manure, ash—no chemical fertilizer
- **Pesticides:** Neem, local botanicals, cultural methods—no external chemicals
- **Crops:** Full diversity: staples (rice/maize/millet), pulses, vegetables, fruits, spices—sufficient for year-round diet
- **Livestock:** Buffalo/cows (milk, manure, draft), goats, chickens, pigs—for protein, eggs, manure
- **Storage:** *Bharako*, traditional methods—no cold chain
- **Processing:** Own rice mill, oil press, flour mill—powered locally
- **Preservation:** Drying, pickling, smoking, fermentation
- **Hard dependency:** Salt (unless local salt source, e.g., Himalayan); iodized salt for health

24.4 Shelter Independence

- Own building materials: stone, mud, wood, bamboo, thatch, slate
- Own masons, carpenters, thatchers
- Own repair and construction capacity
- Earthquake-resistant techniques (local knowledge)

24.5 Clothing & Textile Independence

- Own fiber: cotton, wool, hemp, nettle, silk (if sericulture)
- Own spinning, weaving
- Own tailoring, repair
- Own dye: natural dyes from plants, minerals
- Own footwear: leather (if livestock), rubber-soled (hard dependency)

24.6 Transport Independence (Internal)

- Foot trails, bridges (locally built)
- Pack animals: mules, horses, yaks—bred and maintained locally
- Porters (human labor)
- No need for motor vehicles if no external travel
- Boats/rafts if river within village

24.7 Education Independence

- Own primary school, teachers from village (trained or apprenticed)
- Own secondary school (or sufficient population to justify)
- Own vocational training: apprenticeship within village
- Own library, books (printed—hard dependency; or oral tradition, handwritten)
- Curriculum: state-approved or community-defined
- Own school management, no external funding

24.8 Healthcare Independence

- Own health worker: trained FCHV, midwife, or community health worker
- Own pharmacy: herbal medicine, Ayurveda, basic allopathic (stocked—hard dependency)
- Own birthing capacity: skilled birth attendant
- Own capacity for: wounds, fractures, common illnesses, dental extraction
- Own mental health: community support, traditional healers (*dhami/jhankri*)
- Own vaccination: cold chain impossible without electricity; or accept disease risk
- **Hard dependency:** Antibiotics, vaccines, surgical equipment, complex diagnostics—cannot be produced locally; 100% independence means accepting limits or minimal strategic stockpile from one-time trade

24.9 Governance & Administration Independence

- Own village council, ward committee—elected or customary
- Own dispute resolution: mediation, *panchayat*-style
- Own record-keeping: births, deaths, land—written or oral
- Own enforcement: community norms, local guards
- **State interface:** Citizenship, national ID, land registration—minimal; village can maintain internal records; full independence implies village as de facto autonomous unit (historical *rajya*, *thum*)

24.10 Justice & Security Independence

- Own dispute resolution for civil, family, property
- Own security: no external police; community vigilance
- Own disaster response: search, rescue, relief from local stocks
- Own fire response: bucket brigades, community

24.11 Communication Independence

- Internal: word of mouth, village crier, notice boards
- Own postal runner to neighboring villages (if any exchange needed)
- Own radio transmitter (if electricity and equipment—hard dependency)
- No mobile, internet—or own community network (very advanced)

24.12 Commerce & Finance Independence

- Barter, labor exchange (*parma*, *parma*)
- Own grain bank, seed bank
- Own savings group, rotating fund
- No external currency dependency for internal transactions
- No bank, no remittance—economy is closed

24.13 Sanitation & Waste Independence

- Own toilets: pit, compost—no external septic services
- Own waste: composting, reuse, minimal dumping
- Own drainage: constructed and maintained locally

24.14 Industry, Craft & Repair Independence

- **Blacksmith:** Tools, implements, repair—ore (hard: need local iron or recycling)
- **Carpenter:** Furniture, construction, repair
- **Potter:** Utensils, storage
- **Weaver, tailor:** As above
- **Miller:** Rice, flour, oil
- **Bicycle/motorcycle repair:** Only if vehicles used; parts (hard dependency)
- **Solar/electrical repair:** If solar used; components (hard dependency)

24.15 Raw Materials & Hard Dependencies

- **Salt:** Himalayan rock salt in some areas; otherwise trade
- **Iron/steel:** Mining rare; recycling from scrap; otherwise trade
- **Soap:** Can make from ash + local oil
- **Paper:** Can make from lokta, recycled; or palm leaf, bark
- **Glass:** Cannot produce locally; use oiled paper, open windows
- **Rubber, plastic:** Cannot produce; minimize use
- **Medicines:** Herbal for most; critical allopathic—strategic reserve or accept risk

24.16 Population & Demographics

- Sufficient population to fill all roles: farmers, blacksmith, carpenter, teacher, health worker, etc.
- Marriage pool: internal or arranged with neighboring villages (reduces strict independence)
- Birth rate to sustain population
- Care for elderly, disabled, orphans—within community

24.17 Social, Cultural & Religious

- Own temples, *gumbas*, places of worship
- Own community hall
- Own festivals, rituals, ceremonies

- Own entertainment: music, dance, storytelling
- Own cremation/burial grounds

24.18 Environment & Sustainability

- Community forest: sustainably managed for fuel, timber
- Grazing land for livestock
- Soil conservation, terrace maintenance
- No over-extraction: water, forest, soil

24.19 Time, Calendar & Measurement

- **Timekeeping:** Sundial, water clock, stars, rooster—for work rhythm
- **Calendar:** Lunar (*tithi*), solar—for planting, festivals, rituals
- **Agricultural calendar:** Who knows when to plant? Weather signs, traditional knowledge
- **Measurement:** *Pathi*, *mana*, *dharni* for grain; *haat*, *kosh* for length—standardization within village
- **Value:** How is labor valued in *parma*? How much grain for a day's work? Custom, negotiation

24.20 Surplus, Buffer & Risk

- **Famine reserve:** How many months of grain? One bad year? Two? Strategic storage
- **Winter buffer:** High mountains—6+ months snow; fuel, food, fodder stored
- **Disaster stock:** Earthquake, flood—immediate relief from local stores
- **Crop failure:** Hail, drought, pest—diversity, fallback crops
- **Livestock epidemic:** Quarantine, culling, herbal treatment—no vet

24.21 Geographic & Resource Reality

- **All within boundary:** Arable land, forest, water source, pasture, settlement—must all exist within village territory
- **Altitude:** Hill villages often need land at multiple elevations (rice low, potato high); scattered plots
- **Transhumance:** Seasonal movement of livestock—grazing rights
- **Water source:** Spring, river—must be within or legally secured; upstream villages can cut supply
- **Quarries:** Stone, clay, lime—local deposits or trade

24.22 Skill Redundancy & Succession

- **No single point of failure:** If blacksmith dies, who replaces? Multiple apprentices
- **Knowledge transfer:** Oral, written, apprenticeship—before elder dies
- **Leadership succession:** How is next leader chosen? Hereditary, election, rotation
- **Chicken-egg:** Blacksmith needs anvil; potter needs wheel—initial tools from somewhere; recycling, inheritance

24.23 Death, Ritual & Continuity

- **Cremation/burial:** Wood for pyre, land for burial—who provides?
- **Ritual:** Who performs? Priest, lama—full-time or part-time; how compensated?
- **Inheritance:** Land, house, tools—rules for division; fragmentation over generations
- **Orphan:** Who raises? Kinship, adoption
- **Widow:** Property rights, remarriage—custom

24.24 Conflict, Crime & Punishment

- **Dispute:** Civil, family, property—mediation; what if parties refuse?
- **Crime:** Murder, theft, rape—what punishment? Fine, exile, corporal? No jail without guards, food
- **Factionalism:** Family feud, caste conflict—can split village
- **Boundary dispute:** With neighboring village—grazing, forest, water; no external arbiter

24.25 Defense & External Threat

- **Bandits, raiders:** Historical reality—walls, vigilance, weapons?
- **Wild animals:** Leopard, bear, wild boar—crops, livestock, humans; how to deter?
- **Neighboring village:** Conflict over resources—defense or negotiation

24.26 Healthcare: The Limits

- **Snakebite:** Antivenom—cannot produce; herbal alternatives limited
- **Complicated childbirth:** Breech, hemorrhage—traditional skills can save some; many die
- **Appendicitis, trauma:** Surgery—no surgeon; accept death
- **Cataract, severe dental:** Blindness, chronic pain—camp-based care requires external visit
- **Mental illness:** Severe—restraint, care; no psychiatric drugs
- **Contagious epidemic:** Quarantine, isolation—who enforces?

24.27 Population & Demographics: The Numbers

- **Minimum viable:** Estimate 200–500+ to fill all roles with redundancy
- **Genetic diversity:** Inbreeding if too closed; marriage with neighbors reduces independence
- **Birth/death balance:** Sustain population; family planning—natural methods
- **Disability, chronic illness:** Care within family/community—productivity loss

24.28 Social Obligation & Reciprocity

- **Hospitality:** Cultural duty to feed guests—uses surplus
- **Festival feasts:** Who provides? Rotating, pooling
- **Gift economy:** Marriage, death, festivals—reciprocity, social capital
- **Common labor:** *Shramdan*—trail, canal, temple—who organizes? Who participates?

24.29 Division of Labor: Caste, Gender

- **Traditional:** Blacksmith (*kami*), tailor (*damai*), etc.—caste-based; does independence require breaking or retaining?
- **Gender:** Who fetches water? Who plows? Who weaves? Who cares for children?
- **Child labor:** At what age do children work? School vs. field

24.30 Childcare, Elderly & Disability

- **Childcare:** When parents in field—grandmother, older sibling, communal
- **Elderly:** Bedridden—who feeds, bathes? Family, rotation
- **Disabled:** Born or acquired—who supports? No pension; family, charity

24.31 Storage & Preservation

- **Grain:** Rats, weevils—how to protect? *Bharako* design, neem leaves, smoke—studies report 5–20% cereal storage loss in Nepal; hermetic storage adoption remains low (see References)

- **Seed viability:** Saving year to year—maintain purity, germination

24.32 Pollution & Waste

- **Blacksmith:** Smoke, slag—where?
- **Tannery:** If leather—waste, smell
- **Latrine:** Pit filling—where next? Rotation

24.33 Beauty, Meaning & Spirit

- **Art:** Mural, carving, craft—for meaning, not just function
- **Entertainment:** Music, dance, storytelling—who performs? When?
- **Instruments:** Drum, flute—who makes? Who plays?
- **Why:** Purpose beyond survival—ritual, festival, belonging

24.34 The Hard Truths

1. **100% is impossible** without accepting death, disability, and suffering that modern medicine could prevent.
2. **Geographic luck** matters: a village without local iron, salt, or sufficient water cannot be independent.
3. **Population** must be large enough for all skills plus redundancy—small villages cannot.
4. **Marriage** with neighbors breaks strict independence; inbreeding breaks health.
5. **Surplus** is essential—for bad years, festivals, guests—but surplus requires good land, labor, and luck.
6. **Conflict** without external arbiter can destroy a village from within.
7. **Independence** was historically possible when villages were *rajya*, *thum*—semi-autonomous. The modern state, market, and aspiration have made it rare.

24.35 Summary: Independence vs. Reality

A 100% fully independent village is theoretically possible only if: (1) it produces or harvests everything from local resources, (2) it has sufficient population and skill diversity, (3) it accepts **hard dependencies** for a few items (salt, iron, critical medicines) via minimal trade, or (4) it accepts higher risk (disease, crop failure) without external buffer. In practice, *maximum self-reliance* is the achievable goal—minimizing external dependency while maintaining dignity and resilience.

25 Summary: The Continuum

Fully Remote Local water, firewood, subsistence agriculture, foot/mule transport, health posts, primary schools, ward office, radio, occasional mobile. No grid, no roads, no internet.
Partially Connected Micro-hydro or solar, trail/ropeway, secondary school access, district hospital referral, weekly market, patchy mobile.
Fully Connected Roads, grid, internet, banks, higher education, better healthcare, LPG, vehicles, e-services.
100% Independent Maximum self-reliance: local water, fuel, food, shelter, education, healthcare, governance, industry—minimal external trade for salt, iron, critical medicines.

Understanding this continuum helps in planning interventions, policy, and development projects tailored to each village's reality.

26 References and Further Reading

The following research articles, books, and reports inform this framework. Links and citations are provided for deeper study.

26.1 Infrastructure & Wellbeing

- **Access to infrastructure and human well-being: evidence from rural Nepal.** Examines how roads, water, irrigation, electricity, and other infrastructure affect wellbeing; finds roads highest priority, followed by drinking water and irrigation; remote areas show 50–65% lower per capita income, higher malnutrition and dropout. https://scispace.com/pdf/access-to-infrastructure-and-human-well-being-evidence-from-4uuy9g9dkt.pdf
- **Rural Living in Nepal: Housing, Water & Waste in Paanchkhal.** NPRC Journal of Multidisciplinary Research. Housing (Pakka/Kachcha), water sources, purification, latrines, waste disposal. https://nepjol.info/index.php/nprcjmr/article/view/78324

26.2 Water & Irrigation

- **Does Rural Water System Design Matter? A Study of Productive Use of Water in Rural Nepal.** Water 11(10), 1978. Productive use of water beyond drinking. https://www.mdpi.com/2073-4441/11/10/1978
- **Functionality of Rural Community Water Supply Systems and Collective Action: Guras Rural Municipality, Karnali.** https://nppr.org.np/index.php/journal/article/view/26/53
- **Sulitar Irrigation System: Struggle from Poverty to Prosperity.** Hydro Nepal. Farmer-managed irrigation, participatory approach, community ownership. https://www.nepjol.info/index.php/HN/article/view/13276
- **Farmers in Nepal reap benefits of rehabilitated farmers-managed irrigation systems.** Asian Development Bank. FMIS rehabilitation, crop intensification, income gains. https://www.adb.org/results/farmers-nepal-reap-benefits-rehabilitated-farmers-managed-irrig
- **Improving irrigation governance and management in Nepal.** Virginia Tech. 20+ years of research on community resource management. https://vtechworks.lib.vt.edu/items/cbc1aa78-7c13-4404-bfbf-2552edcc72e7
- **Irrigation in Nepal: Opportunities and Constraints.** Third World Centre. https://thirdworldcentre.org/wp-content/uploads/2020/07/RPP-Dec-1-1989-Irrigation-in-Nepal-Opportun pdf

26.3 Food Security & Post-Harvest

- **Karki TB, Sah SK, Thapa RB, McDonald AJ, Davis AS (2015) Identifying Pathways for Improving Household Food Self-Sufficiency Outcomes in the Hills of Nepal.** PLoS ONE 10(6): e0127513. CART analysis; agronomic management, maize-fingermillet yields, improved cultivars as primary drivers. https://doi.org/10.1371/journal.pone.0127513
- **Post-harvest practices of horticultural crops in Nepal: Issues and management.** 20–50% post-harvest loss; cold storage, packaging. https://www.academia.edu/66093653/Post_harvest_practices_of_horticultural_crops_in_Nepal_Issues_and_management
- **On-Farm Grain Storage and Challenges in Bagmati Province, Nepal.** Sustainability. 5–20% storage loss; pests, monsoon; hermetic storage adoption low. https://www.cgiar.org/research/publication/on-farm-grain-storage-and-challenges-in-bagmati-province-nepal/
- **Assessing the Impact of Hermetic Storage Technology on Post-harvest Losses among Maize Farmers in Nepal.** Preprints. https://www.preprints.org/manuscript/202411.1421/v1

26.4 Energy

- **Techno-Economic Modelling of Micro-Hydropower Mini-Grids in Nepal.** Energies. Financial sustainability, electric cooking, productive use. https://www.mdpi.com/2071-1050/14/16/4232
- **Identifying the best decentralized renewable energy system for rural electrification in Nepal.** Journal of Asian Rural Studies. Micro-hydro ranked best, then solar. http://pasca.unhas.ac.id/ojs/index.php/jars/article/view/2097
- **Performance evaluation of solar PV mini grid system in Nepal: Thabang and Sugarkhal.** Frontiers in Energy Research. https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2023.1321945/full
- **Sustainability assessment of a micro hydropower plant in Nepal.** Energy, Sustainability and Society. https://energsustainsoc.biomedcentral.com/articles/10.1186/s13705-018-0147-2

26.5 Health & Education

- **Perception of Health and Care Seeking Behaviors in High-Altitude Villages of Rural Nepal.** HPROSPECT. Traditional healers, cultural beliefs, accessibility. https://nepjol.info/index.php/HPROSPECT/article/view/24103
- **Assessing the health impacts of a Comprehensive Rural Health Project in Nepal.** PLOS Global Public Health. Village Alive Project, Dalit villages. https://journals.plos.org/globalpublichealth/article?id=10.1371/journal.pgph.0004458

26.6 Rural Development & Transport

- **Rural Transportation Infrastructure in Low- and Middle-Income Countries.** Sustainability. Impacts, interventions, links to agriculture and health. https://www.mdpi.com/2071-1050/14/4/2149
- **Handbook of Rural Development.** Comprehensive rural development scope.
- **Infrastructure for Smart Villages.** Routledge. Fit-for-purpose rural infrastructure, SDG alignment.
- **UN Handbook on Infrastructure Asset Management.** Managing infrastructure for sustainable development.